Literature Status: Question 1 of arXiv:1107.2810

Run `fa_banach_001`, lane 15

2026-06-29

Status

literature_already_answered, Question 1 only.

This packet records that the first subsymmetric-basis question in Kutzarova–Manoussakis–Pelczar-Barwacz [1] is already answered in later literature. It is not an original proof from this run.

Original Question

The source paper is D. Kutzarova, A. Manoussakis, and A. Pelczar-Barwacz, *Isomorphisms and strictly singular operators in mixed Tsirelson spaces*, arXiv:1107.2810; J. Math. Anal. Appl. 388 (2012), no. 2, 1040–1060 [1]. In Section 3.3, "Remarks and questions", on page 27 of the local rendered PDF, it asks:

> Does there exist a space in which all subsymmetric basic sequences are equivalent to one basis, and that basis is not symmetric?

It follows that the operator extending the mapping $y_n \to x_{m_n}$ factors through a c_0-saturated space and hence is strictly singular. $\square$

3.3. **Remarks and questions.** As a corollary to Theorem 3.4, part (1), we obtain that the (non-modified) Tzafriri space Y has an asymptotic ℓ_2 subspace Z which satisfies a blocking principle in the sense of [14]. The only known spaces with a blocking principle so far were similar to T, T^* and their variations. The two major ingredients used in [14] for proving the minimality of T^* are the blocking principle and the saturation with ℓ_∞^n's. It is shown in [21] that Tzafriri space Y contains uniformly ℓ_∞^n's. It is not known whether Y is uniformly saturated with ℓ_∞^n's. In the opposite direction, we do not know if Z contains a convexified Tsirelson space $T^{(2)}$ (which is equivalent to its modified version).

In 1977 Altshuler [2] (cf. e.g. [26]) constructed a Banach space with a symmetric basis which contains no ℓ_p or c_0, and all its symmetric basic sequences are equivalent. In 1981 C. Read [31] constructed a space with, up to equivalence, precisely two symmetric bases. More precisely, Read proved that any symmetric basic sequence in his space CR is equivalent either to the u.v.b. of ℓ_1 or to one of the two symmetric bases of CR. A careful look at the papers of Altshuler and Read shows that their proofs work similarly for the more general case of all subsymmetric basic sequences. This observation leads to the following questions:

Question 1. Does there exist a space in which all subsymmetric basic sequences are equivalent to one basis, and that basis is not symmetric?

We remark that Altshuler's space has a natural subsymmetric version but we do not know if it satisfies the above property.

Question 2. Does there exist a space with exactly two subsymmetric bases, which are not symmetric?

27

Supporting Answer

The supporting paper is S. J. Dilworth, D. Kutzarova, B. Sari, and S. Stankov, *Duals of Tirilman spaces have unique subsymmetric basic sequences*, arXiv:2210.15744 [2]. Its bibliography identifies [1] as the 2012 JMAA version of arXiv:1107.2810. In the introduction it says that the unique non-symmetric subsymmetric basic sequence question posed in [1] and in [3] had been answered by the $Su(T^*)$ construction of Casazza–Dilworth–Kutzarova–Motakis.

The same paper then gives further examples. Its Theorem 10 states that for $1 < p < \infty$ and sufficiently small $\gamma > 0$, every subsymmetric basic sequence in the dual Tirilman space $Ti^*(p, \gamma)$ is equivalent to the canonical basis (e_j^*), and that this canonical basis is subsymmetric but not symmetric.

Why This Answers Question 1

Question 1 asks for a Banach space whose subsymmetric basic sequences all lie in one equivalence class, with that class represented by a non-symmetric basis. Theorem 10 of [2] gives such spaces, namely the duals $Ti^*(p, \gamma)$ for the stated parameter range. The introduction of [2] also explicitly links the question to [1], so this belongs in `literature_already_answered` rather than in `literature_implied_answers`.

Scope Limitations

This packet does not claim an answer to Question 2 of [1], which asks for a space with exactly two subsymmetric bases that are not symmetric. It also does not address the neighboring Tzafriri-space questions about uniform ℓ_∞^n-saturation or containment of a convexified Tsirelson space.

Evidence Checked

The lightweight run indexes had no exact prior entry for arXiv:1107.2810 or for this source question. The local source of arXiv:1107.2810 was inspected around Section 3.3. Exact phrase and keyword searches for the subsymmetric Question 1 surfaced arXiv:2210.15744, and its local source was checked around the abstract, introduction, Theorem 10, and bibliography.

References

[1] D. Kutzarova, A. Manoussakis, and A. Pelczar-Barwacz, *Isomorphisms and strictly singular operators in mixed Tsirelson spaces*, J. Math. Anal. Appl. 388 (2012), no. 2, 1040–1060; arXiv:1107.2810.

[2] S. J. Dilworth, D. Kutzarova, B. Sari, and S. Stankov, *Duals of Tirilman spaces have unique subsymmetric basic sequences*, arXiv:2210.15744.

[3] F. Albiac, J. L. Ansorena, S. J. Dilworth, and D. Kutzarova, *A dichotomy for subsymmetric basic sequences with applications to Garling spaces*, Trans. Amer. Math. Soc. 374 (2021), no. 3, 2079–2106; arXiv:1910.02414.